AI MINI PROJECT

RUNS IN THE BROWSER

ONDRU

ஒன்று · many voices, one issue

On-device AI triage for a municipal grievance desk. It groups re-reports of one incident into a single card, routes it to a desk, ranks what is dangerous, and refuses to guess when it is not sure.

SUBMITTED BY

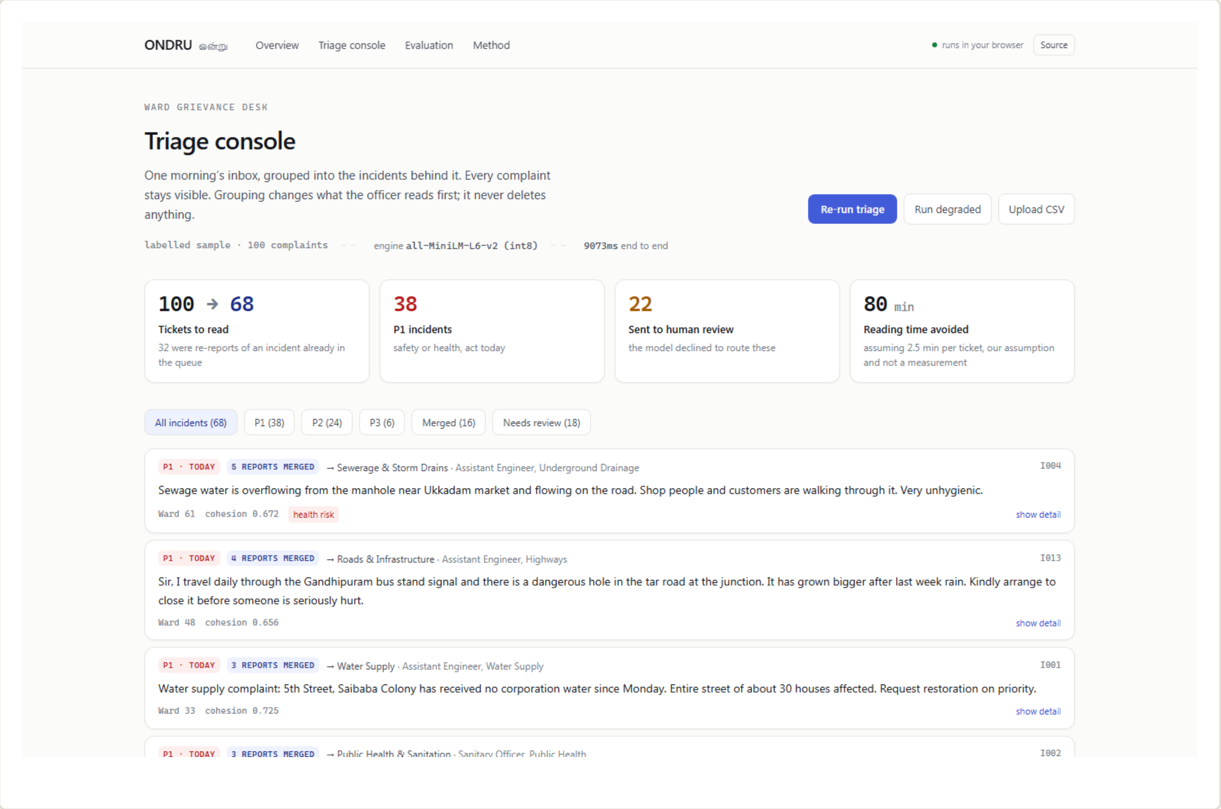
Shyam Prasath S

Sri Krishna College of Technology, Coimbatore

TRACK

SDG 16 · Peace, Justice and Strong Institutions

Code for Communities · Tech for Good 2026



The console. 100 complaints in a ward inbox, grouped into 68 real incidents.

LIVE APPLICATION	SOURCE CODE	DEMO VIDEO
quantum-pi-blush.vercel.app	github.com/ishyamprasath/quantum	drive.google.com/file/d/1wskAFJJ6Vx5H08E7pGRBakwX2vZdtDsc/view

Problem Statement

Why a ward grievance desk is slow, and where the complaints go missing.

THE COST

49%

of the sample queue were re-reports of an incident already open

measured on 100 labelled complaints

200+

complaints a day into one shared ward inbox, sorted by hand

CODERA, team 295, Code for Communities

01

DUPLICATES ARE READ AS NEW WORK

Eleven reports of one pothole become eleven tickets. The crew goes once, one ticket closes, and ten stay open with nobody looking at them.

02

ROUTING IS MANUAL AND SLOW

A clerk reads each complaint to guess a department. A wrong desk does not bounce back. It sits in the wrong queue while the resident is told “under process”.

03

URGENCY IS FIRST COME, NOT SEVERITY

An open manhole waits behind a request to repaint a speed breaker, because the queue is ordered by arrival time and nothing else.

04

AI TRIAGE TOOLS GUESS WHEN UNSURE

A model that always answers will confidently file “please do the needful” to a desk. A confidently wrong label is worse than no label at all.

THE GAP ONDRU FILLS

“Show me the incidents, not the messages, and tell me plainly when you do not know which desk it belongs to.”

Proposed Solution

ONDRU
many voices, one issue

One queue, done properly, rather than a dashboard of four half-features.

INGEST › EMBED › CLUSTER › ROUTE › RANK › HAND BACK

MOAT 01

IT NEVER LEAVES THE TAB

- all-MiniLM-L6-v2 runs in the browser on ONNX Runtime Web. No API key, no server, no complaint text in transit.
- That is the only reason a ward office can point it at a live inbox without a data sharing note.

MOAT 02

IT IS ALLOWED TO SAY NO

- Routing needs a score floor AND a margin over second place. Fail either and nothing is assigned.
- The complaint lands in a visible review lane with the numeric reason. Never dropped, never auto filed.

MOAT 03

EVERY NUMBER IS RECOMPUTED LIVE

- The evaluation page scores 100 hand-labelled complaints in your browser when you press the button.
- Nothing is hard-coded. If the model gets worse, the page gets worse.

WHAT THE DESK GETS

- 100 complaints collapse to 68 incident cards, P1 first.
- Every original report stays attached to its card.
- 22 items handed back with the reason they failed the gate.
- A live metrics page the officer can audit, plus CSV upload for their own inbox.

Features

ONDRU

many voices, one issue

Six capabilities. The fourth is the one most triage demos skip.

FEATURE 01

INCIDENT GROUPING

- Complete-linkage clustering on cosine similarity, not single-linkage.
- A report joins an incident only if it clears the threshold against every member, so unrelated complaints are never chained together and closed as one.

FEATURE 02

DESK ROUTING

- Eight prototype vectors, one per municipal desk, built from the phrasing residents actually use.
- 96.2% accurate on what it answered, across the labelled set.

FEATURE 03

SEVERITY RANKING

- Similarity to three severity prototypes plus a nine-rule hazard lexicon that is printed in full.
- No language model writes the priority, so a supervisor can point at the rule that fired.

FEATURE 04

THE REFUSAL LANE

- Below the floor or inside the margin, nothing is assigned.
- It abstained on 7 of the 7 complaints a human reader could not route either.

FEATURE 05

GRACEFUL DEGRADATION

- If the model cannot be fetched, a deterministic n-gram engine takes over with its own tuned thresholds.
- The desk still gets grouping on a bad connection instead of a spinner.

FEATURE 06

AUDITABLE BY DESIGN

- Every threshold, prototype and lexicon rule is on the method page.
- The sweep that chose them ships in the repository as tune.mts.

Idea and Approach

Every layer is a component that already shipped. Nothing here is speculative.

MODEL

SENTENCE ENCODER

- Xenova/all-MiniLM-L6-v2
- 6-layer MiniLM, int8 quantised
- About 23 MB, cached after first load
- 384-dim, mean-pooled, L2 normalised
- Dot product is the cosine

RUNTIME

BROWSER INFERENCE

- transformers.js v3
- ONNX Runtime Web, WASM backend
- Batched 16 at a time, yields to paint
- No API route, no key, no database
- 100 complaints in about 7 seconds

GROUPING

CLUSTERING

- Complete-linkage agglomerative
- Pairs sorted by similarity, merged greedily
- Merge refused unless every pair clears
- Threshold 0.54, swept not guessed
- Cohesion shown per incident

ROUTING

PROTOTYPE CLASSIFIER

- 8 desks by 5 exemplar phrases
- Prototype is the normalised mean
- Two-part gate: floor 0.35, margin 0.04
- Zero-shot, no labelled training run
- Adding a desk is adding five sentences

SEVERITY

HYBRID SCORING

- 3 severity prototypes, P1 / P2 / P3
- 9-rule signed hazard lexicon, weight x2
- Grid-searched on the labelled set
- 84.6% P1 recall
- Zero true P1 demoted to routine

DELIVERY

SERVING AND UI

- Next.js 16 App Router, React 19
- TypeScript throughout, strict
- Tailwind v4, single light theme
- Static pages on Vercel
- No server-side inference to pay for

System Architecture and Workflow

ONDRU

many voices, one issue

One inbox, from raw messages to a ranked incident queue, entirely in the tab.

NORMALISE › EMBED › CLUSTER › ROUTE › RANK › HAND BACK

Every stage runs inside the browser tab. No API route, no database, no key, no complaint text in transit.

INBOX

WhatsApp forwards
Transliterated Tamil, no punctuation

Portal and email
Formal register, long form

Phone notes
Third-person clerk summaries

ENCODER

all-MiniLM-L6-v2
6-layer transformer, int8, ~23 MB, cached after first load

384-dim, L2 normalised
Mean-pooled, so a dot product is the cosine

Lexical fallback
Hashed n-gram TF, 512-dim, zero download

DECIDE

Complete-linkage cluster
cosine ≥ 0.54 against every member, not just one

8 desk prototypes
Gate: score ≥ 0.35 AND ≥ 0.04 clear of second place

3 severity prototypes
Plus a 9-rule hazard lexicon, printed in full

DESK OUTPUT

Ranked incident queue
68 cards from 100 complaints, P1 first, every report still attached

Needs human review
Gate failed, so the numeric reason is shown. Never dropped, never auto-assigned

Live evaluation page
All metrics recomputed in-tab against 100 labelled complaints

Next.js 16 · App Router | React 19 · TypeScript | transformers.js v3 | ONNX Runtime Web (WASM) | Tailwind v4 | Vercel · static, no server inference

WHY COMPLETE LINKAGE

- Single linkage chains A to C through any middling B.
- Here the Ukkadam and Sungam drain reports would merge.
- One then hides behind the other and closes with it.

ASYMMETRIC ERRORS

- Splitting one incident into two cards wastes thirty seconds.
- Merging two incidents into one loses a complaint.
- So the threshold sits tight of the F1 peak.

Innovation and Existing Solutions

ONDRU

many voices, one issue

What a ward desk can use today, and where ONDRU sits.

CAPABILITY COMPARISON

CAPABILITY	MANUAL SORTING	KEYWORD RULES	CLOUD LLM API	GRIEVANCE PORTALS	ONDRU
Groups duplicate reports	✗	partial	✓	✗	✓
Routes to a department	✗	partial	✓	✓	✓
Ranks by severity, not arrival	✗	✗	✓	✗	✓
Refuses to guess when unsure	n/a	✗	✗	✗	✓
No text leaves the machine	✓	✓	✗	✗	✓
Works with no API key	✓	✓	✗	✗	✓
Published accuracy numbers	✗	✗	✗	✗	✓
Marginal cost per 1,000	8 hrs	free	~₹120	free	**₹0**

POSITIONING

Cloud LLM triage is the obvious build and it is the one a government desk cannot adopt, because it means posting citizen complaints to a third party, per call, forever. ONDRU takes the same job offline.

WHAT IS NEW

- A grievance triage tool with a measured, visible abstention rate.
- Zero marginal cost per complaint, and nothing to procure.

Impact and Future Scope

What it does today, measured, and the honest next step.

DEDUPE

76.3%
Duplicate grouping F1
pairwise over 4,950 complaint pairs

ROUTING

96.2%
Routing accuracy
on the 84% of complaints it answered

REFUSAL

7 / 7
Correctly abstained
on the deliberately unroutable ones

SAFETY

0
True P1 sent to routine
P1 recall 84.6%, never demoted to P3

ROADMAP

P1 · NOW · PROTOTYPE
100 labelled complaints, live metrics page, deployed and public.

P2 · NEXT · REAL LABELS
200 complaints from one Coimbatore ward, labelled by that ward officer.

P3 · +1 MO · TAMIL SCRIPT
Swap in a multilingual encoder so native Tamil, not just transliteration, is handled.

P4 · +2 MO · OFFICER FEEDBACK
A re-route button that logs corrections, so thresholds tune on real disagreement.

P5 · +3 MO · PORTAL IMPORT
Read a CM Helpline or ASKTN CSV export directly, write back grouped incident ids.

P6 · +6 MO · PILOT
One ward, one month, measured against the desk own resolution times.

STATED LIMITS

The 100 complaints were written by us to match the register real ones arrive in, and the desk anchors were written by the same hand, so treat routing accuracy as an upper bound. Urgency errors are lopsided: it over-calls P1 far more often than it under-calls it. Real labels from a real ward are the next step, not more features.

Thank You

ONDRU · many voices, one issue

DEMO VIDEO

[drive.google.com/file/d/1wskAFJJ6Vx5H08E7pGRBakwX2vZdtDsc/view]

3 minute walkthrough

LIVE APPLICATION

[quantum-pi-blush.vercel.app]

runs in your browser

SOURCE CODE

[github.com/ishyamprasath/quantum]

sweep script included

SUBMITTED BY

Shyam Prasath S

Sri Krishna College of Technology

QUESTIONS

"It ranks and groups. It never closes a complaint."

Ask about the clustering choice, the abstention gate, or anything on the evaluation page.